WADE NGUYEN

Data Scientist & AI Engineer · Computer Vision · Data Analytics

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PROFILE

Data scientist and computer-vision engineer with hands-on experience building Python-based ML, computer-vision, and data-analysis systems. Experienced with CNN-, YOLO-, and R-CNN-based detection models, as well as data pipelines using Pandas, NumPy, SQL, and Matplotlib. Currently exploring AI-assisted engineering workflows with Claude Code to accelerate prototyping, debugging, testing, documentation, and code review. Interested in geospatial data, and optimization to real-world earth-observation and satellite analytics problems.

TECHNICAL SKILLS

- **Programming:** Python, SQL.
- **Data & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Excel.
- **ML & CV:** scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, CNN, YOLO, R-CNN, TensorBoard.
- **Time series & geo data:** time-series CV, feature engineering, REST APIs (Open-Meteo, GloFAS).
- **Optimization:** Google OR-Tools (CP-SAT), greedy heuristics.
- **Databases:** SQL, MongoDB.
- **Engineering practices:** Git, GitHub, pytest, debugging, Makefile, technical documentation, Agile/Scrum.
- **AI-assisted development:** Claude Code (plan mode, agentic workflows, code review, test generation), prompt-based problem solving.
- **Collaboration:** Trello, Slack, sprint planning.

LANGUAGES

- English – Good.
- Vietnamese – Native.
- Finnish – Basic level.

CERTIFICATES

- Deep Learning Specialization — DeepLearning.AI / Coursera.
- Advanced Machine Learning Specialization — Coursera.
- Mathematics for Machine Learning Specialization — Imperial College London / Coursera.

PROJECTS

Satellite Task Scheduler Project | Oct 2026 – Present • Link

- Built a Python earth-observation scheduler with battery, memory, slew-time and per-orbit constraints; compared a CP-SAT model with two greedy heuristics on 4 dataset sizes (250–66k opportunities).
- CP-SAT proved optimality up to 2k opportunities in under 0.5 s and served 100% of requests vs 98% for greedy on the 2k opportunities instance.
- Diagnosed CP-SAT stalling at 66k opportunities and fixed it with a complete greedy warm-start hint approach explored with Claude Code; the result is never worse than greedy.

Technologies: Pandas, NumPy, Google OR-Tools (CP-SAT), Git, GitHub, Claude Code.

Multi-location Flood Risk Prediction Project | Oct 2026 – Present • Link

- Built a leakage-safe next-day flood-risk forecasting pipeline for 5 Vietnamese river locations from 5 years of daily rainfall (Open-Meteo) and GloFAS river-discharge data (~9k location-days).
- Benchmarked Logistic Regression, Random Forest and Gradient Boosting against persistence and prior baselines with date-blocked time-series CV; best model reached PR-AUC 0.89 on the 2024 hold-out vs 0.64 for persistence (13% flood days).
- Engineered hydrology features (lags, rolling rainfall, antecedent precipitation index, seasonality); planned, reviewed, and documented with Claude Code.

Technologies: Pandas, NumPy, Matplotlib, pytest, Git, GitHub, Claude Code.

SAR Flood Mapping Project | Nov 2026 • Link

- Built a pipeline that maps flooded areas from Sentinel-1 radar satellite imagery, comparing classical thresholding, Random Forest and U-Net model.
- Doubled IoU of the standard thresholding method (0.33 to 0.68) and tested the model on Bolivia flood, U-Net reach 0.64 vs 0.59 for global Otsu threshold.

Technologies: Python, PyTorch, NumPy, rasterio, pytest, Claude Code.

Facial Emotional Recognition Project — University of Eastern Finland | Nov 2020 – Mar 2021

- Built R-CNN-based models to recognize facial emotions and estimate age; improved accuracy with model ensembling on FER2013 and a small IMDB celebrity dataset.
- Achieved 69.1% accuracy with only 3.4M parameters on a low-end GPU, vs 70.8–72.7% for VGG19, GoogleNet and Inception-v3 (7M–138M parameters).

Technologies: TensorFlow, OpenCV, Keras, PyTorch, TensorBoard, YOLOv3, VGG19, RCNN, SciPy.

Vessel / Cargo Detection Project — Remote internship — GVC Solution JSC | Apr 2019 – May 2020

- Developed YOLOv2-based models to detect vessels and cargo and estimate their dimensions for bridge-passage clearance.

Technologies: TensorFlow, Keras, OpenCV, TensorBoard, Pandas, NumPy, SciPy, YOLOv2, CNN.

Earlier projects: Pedestrian detection with HOG/LBP + SVM on INRIA (Oulu UAS, 2017–2018) · Business data analysis and database design (Saritasa Vietnam internship, 2017–2018) · Customer data analysis and event website (RichRiverPhotography, 2020–2021).

WORK EXPERIENCE

Logistic Employee — Posti Oy | Apr 2024 – Present

- Handle order data, identify and correct database errors, and label barcode orders to support accurate delivery in a fast-paced operation.

Data Analyst & Logistic Employee — DTH-Palvelut Oy | Nov 2021 – Apr 2024

- Built a database and analyzed and visualized customer and worker data to improve operational efficiency.

Technologies: Pandas, NumPy, Excel, SQL, Matplotlib, MongoDB. (Python based)

EDUCATION

Oulu University of Applied Sciences — Bachelor's in Computer Science | 2014 – 2018
Arcada University of Applied Sciences — Big Data Analytics Specialization | 2018 – 2019
University of Eastern Finland — Master's in Information Technology, Computer Vision | 2019 – 2021
Self-study — Coursera / Udemy: ML, deep learning, CNN specializations | 2021 – 2022
Self-Study — Claude Code and agentic AI workflows — Claude Academy | 2026 – Present

TECHNICAL & INTERESTS

- Building and repairing PCs; disassembled a DSLR camera to study its mechanical and optical components.
- Photography, especially symmetrical, minimalism, and modern architecture.
- Traveling.